Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 3

Time Allowed: 30 Minutes.

General Instructions:

Read the following instructions carefully.

1. There are 13 questions in this question paper with internal choice.
2. SECTION A consists of 13 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

Section A

1. Which of the following alcohols will yield the corresponding alkyl chloride on reaction with concentrated HCl at room temperature? **[1]**

a) $CH_3CH_2-\underset{\underset{CH_3}{|}}{C}-CH_2OH$

b) $CH_3CH_2-\underset{\underset{CH_3}{|}}{\overset{CH_3}{\overset{|}{C}}}-OH$

c) $CH_3CH_2-\underset{\underset{CH_3}{|}}{C}-OH$

d) $CH_3CH_2—CH_2—OH$

2. Progesterone is responsible for **[1]**

a) development of secondary male characteristics.

b) development of secondary female characteristics.

c) controlling menstrual cycle.

d) preparing the uterus for implantation of fertilised egg.

3. Alcoholic compounds react: **[1]**

a) both as nucleophiles and electrophiles.

b) None of these

c) only as electrophiles.

d) only as nucleophiles.

4. The following reaction is:

a) Wolff – Kishner reduction

b) catalytic hydrogenation

c) Clemens reduction

d) catalytic ionisation

[1]

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5. The following experimental rate data were obtained for a reaction carried out at 25^oC: $A_{(g)} + B_{(g)} \rightarrow C_{(g)} + D_{(g)}$ **[1]**

Initial $[A_{(g)}]$/mol dm^{-3}	**Initial $[B_{(g)}]$/mol dm^{-3}**	**Initial rate/mol $dm^{-3}s^{-1}$**
3.0×10^{-2}	2.0×10^{-2}	1.89×10^{-4}
3.0×10^{-2}	4.0×10^{-2}	1.89×10^{-4}
6.0×10^{-2}	4.0×10^{-2}	7.56×10^{-4}

What are the orders with respect to $A_{(g)}$ and $B_{(g)}$?

a) Order with respect to $A_{(g)}$ - Zero Order with respect to $B_{(g)}$ - Second

b) Order with respect to $A_{(g)}$ - First Order with respect to $B_{(g)}$ - Zero

c) Order with respect to $A_{(g)}$ - Second Order with respect to $B_{(g)}$ - First

d) Order with respect to $A_{(g)}$ - Second Order with respect to $B_{(g)}$ - Zero

6. Match the item given in Column I with expression given in Column II. **[1]**

Column I	Column II
(a) Osmotic Pressure	(i) $p = K_H.X_B$
(b) Relative lowering of vapour pressure	(ii) $\frac{\Delta P}{P^o_A} = X_B$
(c) Henry Law	(iii) $\Delta T_b = K_b.m$
(d) Elevation in boiling point	(iv) p = iCRT

a) (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i).

b) (a) - (iii), (b) - (i), (c) - (ii), (d) - (iv).

c) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i).

d) (a) - (iv), (b) - (ii), (c) - (i), (d) - (iii).

7. Chlorine reacts with cold and dilute NaOH to give **[1]**

a) NaCIO and $NaClO_3$

b) NaCl and NaCIO

c) NaCl and $NaClO_3$

d) NaCl and $NaClO_4$

8. Which forms protective and non-corrosive oxide layer? **[1]**

a) Ni

b) Cu

c) Zn

d) Cr

9. If the initial concentration is reduced to $\frac{1}{4}^{th}$ in a zero order reaction, then the time taken for half the reaction to complete: **[1]**

a) doubles

b) remains the same

c) increases four times

d) reduces to one-fourth

10. The reagent that can be used to distinguish acetophenone and benzophenone is

A)aqueous $NaHSO_3$

b)I_2 and NaOH

c) Fehling solution

d) 2, 4-dinitrophenyl hydrazine

11. Aspirin is obtained by the acetylation of which of the following compounds? **[1]**

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[1]

a) Phenol

b) Acetyl salicylic acid

c) Salicylaldehyde

d) Salicylic acid

12. Aniline upon heating with conc. HNO_3 and conc. H_2SO_4 mixture gives: **[1]**

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a) o-and p-nitroaniline

b) No reaction

c) The mixture of o, p, and m nitroaniline:

d) o-nitroaniline

13. **Assertion (A):** Primary structure of proteins gives an idea about the conformation of the molecule. **Reason (R):** The differences in chemical and biological properties of various proteins arise due to the difference in primary structure.

a) Both A and R are true and R is the correct explanation of A.

B) A is true but R is false.

C) Both A and R are true but R is not the correct explanation of A.

D) A is false but R is true.

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